

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (BIOLOGY)

CHAPTER 4: WHY DO WE FALL ILL?

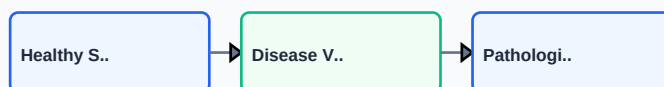
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Biology)
Chapter Name:	Why Do We Fall Ill?

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: HEALTH AND ITS FAILURE



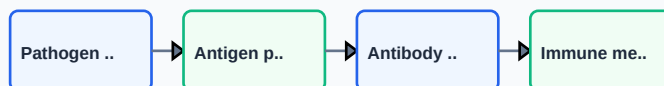
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Health	A state of complete physical, mental, and social well-being, and not merely the absence of disease or infirmity.
Infectious Diseases	Diseases caused by pathogens (microbes) that can spread from an infected person to a healthy individual.
Pathogens	Microscopic disease-causing agents, including bacteria, viruses, protozoa, fungi, and worms.
Immunity	The body's capability to resist and neutralize infections, which can be natural (innate) or acquired.
Vaccination	The administration of weakened or dead pathogens (vaccines) to stimulate the immune system to build antibodies.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

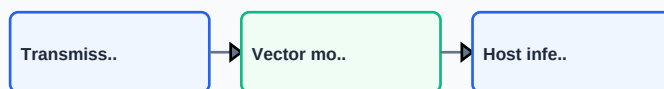
PATHOGEN INFECTION & ACQUIRED IMMUNITY STEPS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: INFECTIOUS DISEASE SPREAD RATE DYNAMICS



Essential Formulas, Laws & Identities:

- Disease transmission methods: Air-borne (Tuberculosis), Water-borne (Cholera), Vector-borne (Malaria), Sexual contact (AIDS).
- Antibiotics: Drugs that block metabolic pathways of bacteria (ineffective against viruses).

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: What is the vector for Malaria and Dengue?

Step-by-step Solution:

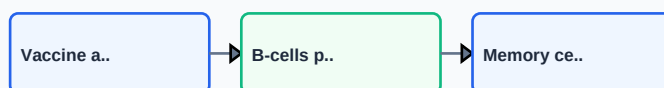
Malaria is transmitted by the female *Anopheles* mosquito. Dengue is transmitted by the female *Aedes* mosquito. The mosquitoes carry the pathogen but do not suffer from the disease.

Example 2: Explain why we cannot cure a common cold with standard antibiotics.

Step-by-step Solution:

Common cold is a viral infection. Antibiotics function by disrupting the biochemical cell wall pathways of bacteria. Since viruses lack cell walls and utilize the host's pathways, antibiotics have no effect on them.

WORKED: VACCINE-INDUCED IMMUNIZATION PROCESS



HOTS (Higher Order Thinking Skills) Challenge

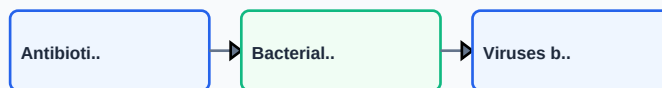
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: How does immunization create long-term protection against severe infections?

Analytical Solution Strategy:

When a vaccine is administered, the immune system produces antibodies and creates 'memory cells'. If the active pathogen attacks in the future, these memory cells immediately recognize it, producing a massive surge of targeted antibodies to destroy the infection before it causes illness.

HOTS: BACTERIAL ANTIBIOTIC RESISTANCE VS VIRAL IMMUNITY



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Explain the difference between acute and chronic diseases with examples.

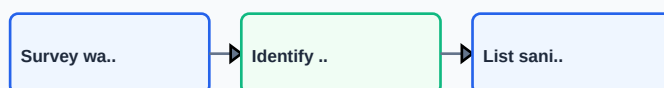
Q2 (2-Mark): State the common symptoms and prevention methods of: (i) Tuberculosis, (ii) Typhoid.

Q3 (2-Mark): How does pulse polio immunization work?

Q4 (5-Mark): Explain the principles of treatment and prevention of infectious diseases.

Q5 (5-Mark): What are vector-borne diseases? How can we control mosquito breeding?

PRACTICE: COMMUNITY HYGIENE INSPECTION CHECKLIST



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Public Health Administration

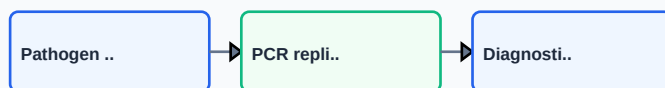
WHO and local health departments design vaccination campaigns (e.g. COVID-19, BCG) to achieve herd immunity.

Application Case: Water Sanitation

Chlorinating public water supplies kills bacterial pathogens, preventing cholera epidemics.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: EPIDEMIOLOGICAL TRACKING OF VIRAL OUTBREAKS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm³, N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: INFECTIOUS DISEASES PREVENTION LOOP

